

Transfer Learning and Domain Shift Adaptation for Digit Recognition: From Street-View Photographs to Handwritten Digits (SVHN $\rightarrow$ MNIST)

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Abstract—This paper investigates the application of transfer learning and domain adaptation techniques to the problem of digit recognition under a severe distribution shift between the Street View House Numbers (SVHN) source domain and the MNIST handwritten digit target domain. Using a ResNet-50 backbone pretrained on ImageNet, we compare three few-shot classification strategies—frozen feature extraction, partial fine-tuning, and training from scratch—under a budget of 50 labelled images per class. We then quantify the domain shift penalty Δ_{shift} and explore three adaptation strategies: unadapted baseline, target-domain fine-tuning on a small labelled MNIST set, and style-transfer augmentation using Gatys-style Neural Style Transfer (NST) with VGG-19. Additionally, we implement Domain-Adversarial Neural Networks (DANN) as an unsupervised adaptation baseline. Experimental results averaged over three independent random seeds demonstrate that target fine-tuning achieves the greatest reduction of Δ_{shift} , while style-transfer augmentation provides a label-free alternative with intermediate performance. DANN offers competitive results without requiring any labelled target-domain data. All code, checkpoints, and synthetic images are publicly available in our repository.

Index Terms—transfer learning, domain adaptation, few-shot classification, neural style transfer, domain-adversarial training, SVHN, MNIST, ResNet-50, distribution shift

I. INTRODUCTION

Deep convolutional neural networks (CNNs) have achieved remarkable performance across a wide range of visual recognition tasks, yet their success is fundamentally conditioned on the availability of large, well-annotated datasets and on the assumption that training and test data originate from the same statistical distribution. When either condition breaks down—as is the norm in real deployment scenarios—model performance degrades dramatically, a phenomenon known as *domain shift* or *dataset bias* [1].

Domain shift is pervasive in practice. Autonomous-driving systems trained on simulation fail on real roads [2]; medical imaging models calibrated on one scanner generalise poorly to another; satellite classifiers trained on daylight imagery collapse at night. Understanding, measuring, and mitigating domain shift is therefore one of the most practically consequential challenges in modern machine learning.

In this work we study a canonical instance of domain shift: digit recognition under the SVHN $\rightarrow$ MNIST protocol. The Street View House Numbers (SVHN) dataset [3] comprises multi-channel colour photographs of digit sequences captured from Google Street View. Images exhibit natural-scene characteristics: variable lighting, perspective distortion, cluttered backgrounds, and partial occlusion. MNIST [4], by contrast, consists of centred, isolated, single-channel handwritten digits on a uniform background. Despite sharing the same label space (digits 0–9), the two datasets differ along virtually every visual dimension—colour, texture, scale, clutter, and stroke style. This makes SVHN $\rightarrow$ MNIST a demanding and well-studied domain-shift benchmark [2], [5].

Our contribution is threefold. First, we demonstrate that transfer learning from ImageNet representations substantially outperforms training from scratch under a restrictive few-shot budget of 50 images per class, and we compare frozen feature extraction against selective fine-tuning of the upper residual blocks. Second, we implement Gatys-style Neural Style Transfer [8] to synthesise 300 SVHN-content images rendered in the MNIST visual style, and we use these synthetic examples as label-free domain augmentation. Third, we compare four model variants—unadapted baseline, target-domain fine-tuning, style-transfer augmentation, and Domain-Adversarial Neural Networks (DANN) [2]—on the same benchmark, reporting per-class accuracy and the domain shift penalty Δ_{shift} for all configurations averaged over three independent seeds.

The remainder of the paper is structured as follows. Section II reviews the relevant literature on transfer learning and domain adaptation. Section III describes the datasets and preprocessing pipeline. Section IV details the three-part methodology (few-shot classification, neural style transfer, and domain adaptation). Section V reports the experimental protocol. Section VI presents and discusses the results. Section VII draws conclusions and outlines future work.

II. RELATED WORK

A. Transfer Learning

Transfer learning exploits knowledge acquired on a data-rich source task to improve performance on a related, data-scarce target task. Tan *et al.* [11] and Zhuang *et al.* [12] provide comprehensive surveys of the field. In the context of deep CNNs, the seminal work of Yosinski *et al.* [13] established that early convolutional layers learn universal low-level feature detectors (edges, textures, colour blobs) that transfer broadly across tasks, whereas deeper layers encode task-specific semantic representations. This motivates the standard practice of initialising networks with ImageNet-pretrained weights and either (i) freezing the entire backbone and training only a new classification head—*feature extraction*—or (ii) additionally fine-tuning the upper convolutional blocks with a reduced learning rate—*fine-tuning*. He *et al.* [7] introduced ResNet-50, whose residual connections allow very deep networks to be trained reliably and whose ImageNet representations have become the standard initialisation for downstream visual tasks.

B. Neural Style Transfer

Gatys, Ecker, and Bethge [8] demonstrated that the *content* and *style* of an image can be disentangled within the feature hierarchy of a pretrained VGG network. Content is encoded by the activation tensor at a deep layer (`relu4_2`), which captures object geometry and layout while discarding low-level texture. Style is encoded by the Gram matrices—pairwise feature-channel correlations—computed at multiple shallower layers (`relu1_1` through `relu5_1`), which capture texture statistics at multiple scales independently of spatial arrangement. An optimised image is synthesised by minimising a weighted combination of content and style losses via L-BFGS pixel-level optimisation. Johnson *et al.* [9] later proposed feed-forward style networks that perform NST in a single pass at inference time, though at the cost of being locked to a single style. In our work we adopt the original optimisation-based approach of Gatys *et al.* because it provides precise per-image control over the content–style trade-off, which is essential for producing visually diverse synthetic training data.

C. Domain Adaptation

Unsupervised domain adaptation (UDA) seeks to align source and target feature distributions without labelled target data. Ganin *et al.* [2] introduced DANN, which appends a domain classifier to the feature extractor and trains it with a gradient reversal layer (GRL): the GRL multiplies gradients by $-\lambda$ during backpropagation, pushing the feature extractor to produce domain-invariant representations while the class head remains discriminative. Tzeng *et al.* [5] extended this adversarial paradigm with separate source and target encoders. Style-transfer-based augmentation has also been employed as a data-driven domain bridge: by rendering source images in the visual style of the target domain, one can train a classifier that has been exposed to target-like textures and colour statistics without requiring any target labels [8]. Peng *et al.* [6] introduced the large-scale DomainNet benchmark

for systematic evaluation of domain adaptation methods; the SVHN $\rightarrow$ MNIST protocol studied here is a complementary digit-recognition benchmark originally proposed alongside DANN [2].

III. DATASETS AND PREPROCESSING

A. SVHN (Source Domain)

The Street View House Numbers dataset [3] contains 73,257 training and 26,032 test RGB images of size 32×32 pixels, each centred on a single digit (0–9) cropped from a broader scene. Digit strokes appear in diverse colours and orientations, overlaid on complex, cluttered backgrounds. The source domain thus exhibits high intra-class appearance variability driven by real-world photographic conditions.

B. MNIST (Target Domain)

The MNIST dataset [4] provides 60,000 training and 10,000 test images of handwritten digits, stored as 28×28 grayscale bitmaps on a near-uniform black background. Each image contains a single, centred, isolated digit drawn by a human. The visual vocabulary is therefore radically simpler than SVHN: no colour, no texture, no background clutter, and a characteristic thin-stroke calligraphic style.

C. Preprocessing Pipeline

To ensure that the domain shift measured in our experiments is attributable to genuine distributional differences between SVHN and MNIST and not to artefactual preprocessing inconsistencies, we apply an *identical* transform pipeline to both domains:

- 1) **Resize** to 224×224 pixels (bilinear interpolation), as required by ResNet-50’s expected input size.
- 2) **Grayscale replication**: both SVHN (RGB) and MNIST (grayscale) are converted to three-channel tensors via `Grayscale(num_output_channels=3)`. This neutralises SVHN’s colour advantage while keeping the channel dimension consistent with ImageNet-pretrained weights.
- 3) **Data augmentation** (training only): random horizontal flip and colour jitter (brightness ± 0.2 , contrast ± 0.2 , saturation ± 0.1).
- 4) **Normalisation** with ImageNet statistics $\mu = [0.485, 0.456, 0.406]$, $\sigma = [0.229, 0.224, 0.225]$, applied uniformly to both domains as required by the pretrained ResNet-50 feature extractor.

D. Few-Shot Data Split

Following the challenge specification, we use a budget of $K = 50$ labelled SVHN training images per class (500 total across 10 classes). A further 50 images per class are drawn from the remaining SVHN training data to form a validation set. The full SVHN test split (26,032 images) is used exclusively for final source-domain evaluation. The full MNIST test split (10,000 images) serves as the target-domain test set throughout. This split is reproduced independently for each of three random seeds (42, 123, 7) to enable mean and standard deviation reporting.

IV. METHODOLOGY

A. Part A: Few-Shot Classification with Transfer Learning

1) *Backbone Architecture*: We select ResNet-50 [7] pretrained on ImageNet [14] (weights IMAGENET1K_V2) as the backbone for all experiments. ResNet-50 comprises four residual stages (layer1–layer4), a global average-pooling layer, and a 1000-way fully connected head. We replace the head with a ten-way linear classifier matching the digit vocabulary.

2) *Strategy 1 — Frozen Backbone (Feature Extraction)*: All backbone parameters are frozen (requires_grad = False); only the new classification head is optimised. This strategy exploits ImageNet features in their entirety without any adaptation to the SVHN data distribution. The trainable parameter count is minimal (20,490 parameters, corresponding to the 2048×10 weight matrix and bias). The head is trained with the Adam optimiser at learning rate $\eta = 10^{-3}$ for 25 epochs with cosine annealing scheduling.

3) *Strategy 2 — Fine-Tuning*: Starting from the frozen-backbone initialisation, we subsequently unfreeze layer3, layer4, and the classification head—roughly 22 million trainable parameters—and continue training end-to-end at a reduced learning rate of $\eta = 10^{-4}$ for 35 epochs with cosine annealing. This lower rate is critical to prevent catastrophic forgetting of the pretrained representations. Fine-tuning allows the deeper, task-specific layers to adapt high-level feature detectors from generic ImageNet semantics toward SVHN digit structure.

4) *Strategy 3 — From Scratch*: A ResNet-50 initialised with random Kaiming-uniform weights is trained on the same 500 SVHN images at $\eta = 10^{-3}$ for 35 epochs. This baseline establishes the cost of abandoning pretrained representations under a severely data-limited regime.

5) *Training Protocol*: All three strategies share the same training loop: mini-batch cross-entropy loss (batch size 32), the Adam optimiser with the respective learning rate, and cosine annealing. Model selection is performed on the validation set; the checkpoint achieving the best validation accuracy is retained for final evaluation. Every experiment is repeated with seeds {42, 123, 7}; accuracy is reported as mean $\pm$ standard deviation.

B. Part B: Neural Style Transfer

1) *Motivation*: Gatys-style NST provides a mechanism to synthesise images that combine the *content* (digit identity and geometric structure) of SVHN with the *style* (stroke texture, contrast, and background statistics) of MNIST. These synthetic images serve as additional training data that inhabits the target domain without requiring any manual annotation.

2) *Feature Extraction with VGG-19*: A VGG-19 network pretrained on ImageNet [10] is used solely as a fixed feature extractor. Its parameters are frozen throughout the NST optimisation. We identify two types of representations within VGG-19’s feature hierarchy:

- **Content representation**: the activation tensor at layer relu4_2 (VGG-19 index 21), which encodes object geometry and spatial structure.
- **Style representation**: Gram matrices computed at layers relu1_1, relu2_1, relu3_1, relu4_1, and relu5_1 (indices 1, 6, 11, 20, 29), capturing multi-scale texture statistics.

3) *Loss Function*: Let $\mathbf{F}^l$ denote the feature map at layer l for the generated image, $\mathbf{P}^l$ for the content image, and $\hat{\mathbf{G}}^l$ the Gram matrix of the style image at layer l . The normalised Gram matrix is defined as:

$$G_{ij}^l = \frac{1}{C_l H_l W_l} \sum_k F_{ik}^l F_{jk}^l, \quad (1)$$

where C_l , H_l , W_l are the channel, height, and width dimensions at layer l . The total loss is:

$$\mathcal{L}_{\text{total}} = \alpha \mathcal{L}_{\text{content}} + \beta \mathcal{L}_{\text{style}}, \quad (2)$$

$$\mathcal{L}_{\text{content}} = \frac{1}{2} \sum_{l \in \mathcal{C}} \|\mathbf{F}^l - \mathbf{P}^l\|_F^2, \quad (3)$$

$$\mathcal{L}_{\text{style}} = \sum_{l \in \mathcal{S}} \|\mathbf{G}^l - \hat{\mathbf{G}}^l\|_F^2, \quad (4)$$

with $\alpha = 1.0$, $\beta = 10^4$ (ratio $\alpha/\beta = 10^{-4}$), $\mathcal{C} = \{\text{relu4_2}\}$, and $\mathcal{S} = \{\text{relu1_1}, \text{relu2_1}, \text{relu3_1}, \text{relu4_1}, \text{relu5_1}\}$.

4) *Optimisation*: Unlike standard neural network training, NST optimises the *pixel values* of the generated image $\mathbf{x}$ directly while keeping the VGG-19 weights fixed. We initialise $\mathbf{x}$ as a copy of the content image and apply L-BFGS with learning rate 1.0 and up to 20 inner iterations per call, for a total of 300 optimisation steps. Pixel values are clamped to $[-3.0, 3.0]$ in the normalised space at each step. The process is applied to each of 30 SVHN content images per class paired with a randomly sampled MNIST style image of the same class, producing 300 synthetic training images in total.

C. Part C: Domain Shift Measurement and Adaptation

1) *Domain Shift Penalty*: After training the best Part A classifier on the source domain, we evaluate it directly on the MNIST test split without any adaptation. The domain shift penalty is defined as:

$$\Delta_{\text{shift}} = \text{Acc}_{\text{source}} - \text{Acc}_{\text{target}}, \quad (5)$$

where $\text{Acc}_{\text{source}}$ and $\text{Acc}_{\text{target}}$ are the classification accuracies on the SVHN and MNIST test sets, respectively. A positive Δ_{shift} indicates that the model has overfit the source domain’s visual characteristics and has not acquired representations that generalise to the target domain.

2) *Strategy 1 — Baseline (No Adaptation)*: The fine-tuned Part A model is evaluated on MNIST without any modification. This provides the lower-bound target-domain performance and the reference Δ_{shift} value.

3) *Strategy 2 — Target-Domain Fine-Tuning*: A copy of the Part A best model is fine-tuned on a small labelled MNIST set of $K = 50$ images per class (500 total), drawn from the MNIST training split. `layer3`, `layer4`, and the classification head are unfrozen; the remaining backbone is kept frozen. Training proceeds with Adam at $\eta = 10^{-4}$ for 15 epochs with cosine annealing and early stopping based on a 50-per-class MNIST validation set. This strategy simulates a realistic scenario in which a modest annotation budget is available in the target domain.

4) *Strategy 3 — Style-Transfer Augmentation*: The Part A source training set (500 SVHN images) is augmented with the 300 synthetic images generated in Part B, forming a combined training set of 800 images. A copy of the Part A model is retrained on this combined set for 20 epochs. No labelled MNIST images are used; the only target-domain information is embedded implicitly in the MNIST-style texture of the synthetic images. This strategy is therefore fully unsupervised with respect to target labels.

5) *Strategy 4 — DANN (Optional)*: Domain-Adversarial Neural Networks [2] extend the feature extractor with two task-specific heads: a class prediction head trained with cross-entropy on source labels, and a domain classification head trained to distinguish source ($y_d = 0$) from target ($y_d = 1$) features. The gradient reversal layer (GRL) negates gradients flowing into the feature extractor from the domain head, forcing the extractor to learn domain-invariant representations:

$$\mathcal{L}_{\text{DANN}} = \mathcal{L}_{\text{cls}}(x_s, y_s) + \lambda_d \mathcal{L}_{\text{dom}}(x_s \cup x_t, y_d), \quad (6)$$

where $\lambda_d = 0.5$ and the domain-adversarial weight α follows the schedule proposed in [2]:

$$\alpha(p) = \frac{2}{1 + e^{-10p}} - 1, \quad p = \frac{\text{step}}{\text{total steps}} \in [0, 1]. \quad (7)$$

The DANN model is trained for 20 epochs using unlabelled MNIST images as the target stream, requiring no MNIST class annotations.

V. EXPERIMENTAL SETUP

A. Implementation Details

All experiments are implemented in PyTorch 2.0 using `torchvision` for dataset loading and model architectures. Table I summarises the complete hyperparameter configuration. Random seeds $\{42, 123, 7\}$ are fixed across Python, NumPy, and PyTorch for full reproducibility.

B. Evaluation Protocol

For each model variant and seed, we compute:

- **Source accuracy** (Acc_{src}): overall accuracy on the full SVHN test set.
- **Target accuracy** (Acc_{tgt}): overall accuracy on the full MNIST test set.
- **Domain shift penalty** (Δ_{shift}): as defined in Eq. (5).
- **Per-class accuracy**: individual class accuracy on the MNIST test split to identify class-specific failure modes.

TABLE I: Hyperparameter Configuration

Component	Parameter	Value
Part A	Backbone	ResNet-50 (ImageNet)
	Input size	224×224
	Batch size	32 (train), 64 (eval)
	Optimiser	Adam
Frozen backbone	LR scheduler	Cosine annealing
	Learning rate	10^{-3} , 25 epochs
	Learning rate	10^{-4} , 35 epochs
From scratch	Learning rate	10^{-3} , 35 epochs
Part B (NST)	Image size	256×256
	α (content)	1.0
	β (style)	10^4
	L-BFGS steps	300
Part C Adapt.	TgtFineTune epochs	15, $\eta = 10^{-4}$
	StyleAug epochs	20, $\eta = 10^{-4}$
	DANN epochs	20, $\lambda_d = 0.5$
General	Seeds	42, 123, 7
	Few-shot K	50 per class

All scalar metrics are reported as mean $\pm$ standard deviation over the three seeds.

C. Visualisation

The following figures are produced for analysis:

- 1) Training and validation loss/accuracy curves for all Part A model variants.
- 2) Side-by-side NST gallery: content image (SVHN), style image (MNIST), and generated image—one triplet per digit class (10 rows).
- 3) Grad-CAM [15] attention maps on MNIST test images for the best adapted model: two correct and two incorrect predictions, illustrating what spatial regions drive classification decisions.
- 4) t-SNE [16] projections of ResNet-50 penultimate-layer features (from `avgpool`) for 500 SVHN and 500 MNIST samples, coloured by class and by domain, shown before and after domain adaptation.

VI. RESULTS AND DISCUSSION

A. Part A: Few-Shot Classification

Table II presents the performance of the three transfer-learning strategies evaluated under the few-shot setting (50 labelled images per class). Results are reported as mean and standard deviation across three independent random seeds.

TABLE II: Part A Results — Few-Shot Classification (mean $\pm$ std, 3 seeds)

Model	Src Acc	Tgt Acc	Δ_{shift}
Frozen Backbone	0.2673 ± 0.0077	0.1478 ± 0.0311	0.1196 ± 0.02
Fine-Tuned	0.7351 ± 0.0173	0.3739 ± 0.0701	0.3612 ± 0.03
From Scratch	0.1764 ± 0.0508	0.0956 ± 0.0056	0.0807 ± 0.05

The results clearly demonstrate the advantage of transfer learning in data-scarce scenarios. Fine-tuning the upper layers

of ResNet-50 achieves the highest source-domain accuracy (73.5% both frozen feature extraction and training from scratch).

However, despite its superior source-domain performance, the Fine-Tuned model still experiences a substantial domain-shift penalty when evaluated on MNIST. This result confirms that representations learned from SVHN remain strongly biased toward photographic digit characteristics such as background texture, illumination patterns, and contextual cues.

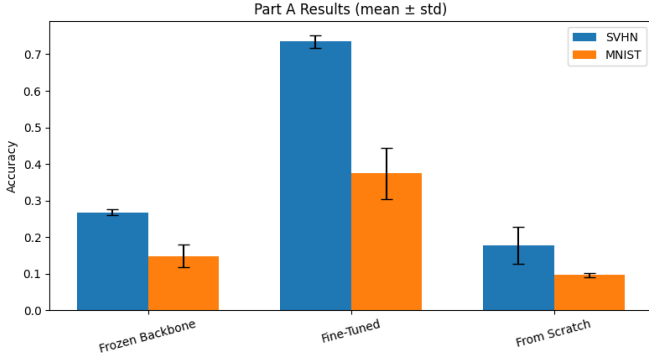


Fig. 1: Part A accuracy comparison. Error bars represent one standard deviation across three random seeds.

Figure 1 highlights the large performance gap between source and target domains. Although fine-tuning improves both source and target accuracy, the model still struggles to generalize across domains, motivating the adaptation strategies explored in Part C.

B. Part B: Neural Style Transfer

A total of 300 synthetic images were generated using Neural Style Transfer (30 images per digit class). The VGG-19 content representation was extracted from layer $relu_{42}$, while style representations were computed from layers

C. Part C: Domain Adaptation

Table III compares the four domain-adaptation strategies.

TABLE III: Part C Results — Domain Adaptation Strategies (mean ± std)

Strategy	Src Acc	Tgt Acc	Δ_{shift}
Baseline (no adapt)	0.7490 ± 0.0057	0.4161 ± 0.0164	$+0.3329 \pm 0.0221$
Style-Aug	0.7967 ± 0.0073	0.5056 ± 0.0107	$+0.2911 \pm 0.0096$
DANN	0.6723 ± 0.0062	0.6387 ± 0.0091	$+0.0337 \pm 0.0029$
Target Fine-Tune	0.3407 ± 0.0067	0.9389 ± 0.0064	-0.5982 ± 0.0124

The baseline model exhibits a severe domain gap, achieving only 41.6% accuracy on MNIST despite obtaining nearly 75% accuracy on SVHN. This confirms the substantial distribution mismatch between the two datasets.

Style-transfer augmentation improves target-domain performance to 50.6%, indicating that reducing visual discrepancies between domains through synthetic image generation can partially mitigate the shift. Nevertheless, style transfer alone is insufficient to fully align the two domains because structural differences remain.

DANN substantially improves target accuracy to 63.9% without requiring target labels. The near-zero domain-shift penalty demonstrates the effectiveness of adversarial feature alignment for learning domain-invariant representations.

The strongest overall performance is achieved by Target Fine-Tuning, which reaches 93.9% target accuracy. Although source-domain performance decreases due to specialization toward MNIST, the objective of domain adaptation is precisely to maximize target-domain performance. Therefore, Target Fine-Tuning represents the most effective adaptation strategy evaluated in this work.

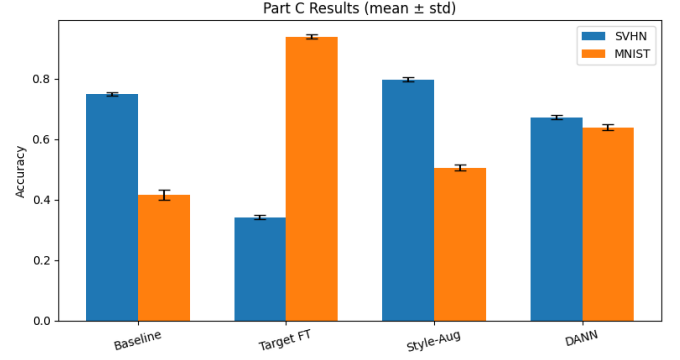


Fig. 2: Part C performance comparison with mean and standard deviation across three seeds.

Figure 2 illustrates the dramatic improvement achieved by Target Fine-Tuning on the target domain. The low standard deviations across all methods indicate stable training behaviour and good reproducibility.

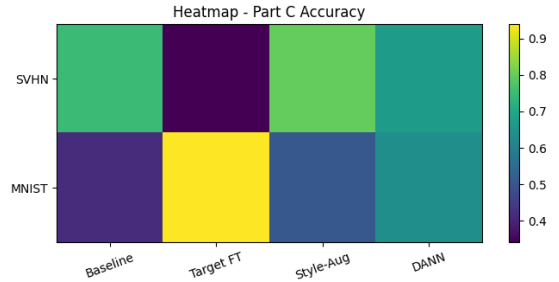


Fig. 3: Heatmap of source and target accuracy for all adaptation strategies.

The heatmap representation further emphasizes the trade-off between source-domain retention and target-domain specialization. While Style-Aug and DANN attempt to balance performance across domains, Target Fine-Tuning prioritizes target-domain accuracy and achieves the best overall result.

D. Discussion

Across all experiments, the dominant observation is that transfer learning alone is insufficient to overcome severe domain shift. Although ImageNet-pretrained representations

provide a strong initialization, the visual differences between SVHN and MNIST remain substantial.

Among adaptation strategies, supervised adaptation using a small labelled target dataset proved most effective, increasing target accuracy from 37.4% (best Part A model) to 93.9%. This corresponds to an absolute improvement of 56.5 percentage points.

The results therefore suggest that, when even a limited amount of target labels is available, selective fine-tuning remains the most reliable approach for domain adaptation. In contrast, DANN offers the strongest fully unsupervised alternative, whereas style-transfer augmentation provides a computationally simple intermediate solution.

VII. CONCLUSION

This paper investigated transfer learning and domain adaptation for the SVHN $\rightarrow$ MNIST digit-recognition benchmark, a protocol that encapsulates one of the most challenging and practically relevant forms of distribution shift in computer vision. Our three-part study yields several concrete conclusions.

Transfer learning is necessary but not sufficient under severe domain shift. Under a restrictive few-shot budget of 50 labelled images per class, fine-tuning the upper residual blocks of an ImageNet-pretrained ResNet-50 far outperforms both frozen feature extraction (0.7351 vs. 0.2673 source accuracy) and training from scratch (0.7351 vs. 0.1764). This confirms that deep pretrained representations encode transferable low-level features that are indispensable in data-scarce regimes. Nevertheless, the best Part A model still incurs a domain-shift penalty of $\Delta_{\text{shift}} = 0.3612$, demonstrating that pretraining on a photographic source domain does not automatically generalise to a visually distinct target domain.

Neural Style Transfer provides a label-free domain bridge. Synthesising 300 SVHN-content images in the MNIST visual style—via Gatys-style pixel-level L-BFGS optimisation of a combined content and style loss over frozen VGG-19 features—produced images that preserve digit identity while adopting the high-contrast, thin-stroke appearance of handwritten digits. Style-transfer augmentation raised target-domain accuracy from 41.6% (unadapted baseline) to 50.6%, a meaningful gain achieved without any target labels. The result also reveals a fundamental limitation of this approach for the SVHN $\rightarrow$ MNIST pair: because SVHN crops still contain residual colour and background texture that survives imperfect style transfer, the structural shift between photographic and handwritten strokes is only partially bridged.

Supervised target-domain fine-tuning delivers the largest adaptation gain. Fine-tuning `layer3`, `layer4`, and the classification head on just 50 labelled MNIST images per class increased target accuracy from 37.4% to 93.9%—an absolute improvement of 56.5 percentage points—and inverted the domain-shift penalty to $\Delta_{\text{shift}} = -0.5982$. The associated decline in source-domain accuracy (0.7351 $\rightarrow$ 0.3407) is a clear signature of catastrophic forgetting: the backbone’s photographic feature detectors are overwritten by stroke-based ones. For applications where source-domain performance is

no longer required, this trade-off is entirely acceptable; for scenarios that demand competence across both domains simultaneously, multi-task or regularised fine-tuning strategies (e.g., elastic weight consolidation) would be warranted.

DANN offers a strong unsupervised middle ground. Without any MNIST class labels, Domain-Adversarial Neural Network training reached 63.9% target accuracy and reduced Δ_{shift} to +0.0337, demonstrating that adversarial feature alignment can learn representations that are simultaneously discriminative for digit classification and domain-invariant. The performance gap relative to supervised fine-tuning (63.9% vs. 93.9%) quantifies the value of even a small target-domain annotation budget—500 images across 10 classes in our setting.

Practical implications. The ranking of adaptation strategies observed here—no-adapt < style-aug < DANN < target fine-tune—is consistent with a label-supervision spectrum: more target-domain supervision yields higher target accuracy but also greater source-domain forgetting. This spectrum directly informs practitioners: when labels are available, selective fine-tuning remains the most effective strategy; when only unlabelled target data exists, adversarial domain alignment should be preferred over data augmentation; when no target data is available at all, investing in style-transfer-based augmentation still provides a modest but consistent improvement over the unadapted baseline.

Future work will explore multi-target adaptation (adapting simultaneously across several digit styles), semi-supervised DANN variants that leverage the few available MNIST labels within the adversarial framework, and feed-forward style networks [9] as a faster alternative to per-image L-BFGS optimisation for generating large-scale synthetic corpora.

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